



## Review

## The effectiveness of psychological interventions for loneliness: A systematic review and meta-analysis



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## ABSTRACT

Chronic loneliness is associated with a range of mental health difficulties. Previous theory and research indicate that psychological interventions show promise for reducing loneliness, however, there have been no systematic reviews or meta-analyses to ascertain the efficacy of these interventions across the lifespan. The aim of this study was to synthesise, meta-analyse and explore the heterogeneity in RCTs of psychological interventions for loneliness in order to establish their efficacy.

Five databases (Ovid Embase, Ovid Medline, PsycINFO, Web of Science and CINAHL) were systematically searched in order to identify relevant studies. Included studies were required to be peer-reviewed RCTs examining psychological interventions for loneliness. Two independent coders examined the abstracts of the 3973 studies and 103 full texts, finding 31 studies that met inclusion criteria, 28 of which contained sufficient statistical information to be included in the meta-analysis. The quality of included studies was assessed using the Cochrane Risk of Bias Tool.

The 31 studies ( $N = 3959$ ) that were included in the systematic review were conducted with participants from a diverse range of cultures, age groups and populations. The interventions were of mixed quality and were mostly face to face, group-based and delivered weekly. The most common type of intervention was Cognitive Behavioural Therapy (CBT).

28 studies ( $N = 3039$ ) were included in a meta-analysis which found that psychological interventions significantly reduced loneliness compared to control groups, yielding a small to medium effect size ( $g = 0.43$ ). Subgroup analysis and meta-regressions were conducted in order to explore heterogeneity and found that type of psychological intervention was approaching significance as a moderator of the effectiveness of psychological interventions for loneliness.

In conclusion, psychological interventions for loneliness across the lifespan are effective. This finding should inform policy makers, researchers and clinicians going forward, especially in the context of increased loneliness due to the COVID-19 pandemic. There was considerable heterogeneity in the effectiveness of the interventions, suggesting that future research should also explore what works for whom and consider personalising psychological treatment.

## 1. Introduction

Loneliness has been defined as a distressing feeling that occurs when there is a discrepancy between desired and achieved social interaction (Peplau & Perlman, 1982), with the importance of subjective perception in this definition making the concept inherently psychological. Loneliness is often thought of as being synonymous with social isolation – an

objective lack of social contact – although research suggests that in fact these related concepts can have an independent relationship with mental health difficulties, such as depression (Ge, Yap, Ong, & Heng, 2017). Loneliness is influenced by both the quantity of social contact and the perceived quality and features of social relations, such as intimacy and trust (Schwarzbach, Luppá, Forstmeier, König, & Riedel-Heller, 2014; Yanguas, Pinazo-Henandis, & Tarazona-Santabalbina, 2018).

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Furthermore, research has shown that loneliness and social isolation have distinct impacts on physical health and mortality (Stephens, Shankar, Demakakos, & Wardle, 2013; Tanskanen & Anttila, 2016).

Loneliness is prevalent and the recent widespread implementation of disease control measures such as social distancing, lockdown and quarantine measures to tackle the recent coronavirus (COVID-19) pandemic have been found to increase rates of loneliness (Banerjee & Rai, 2020; Hwang, Rabheru, Peisah, Reichman, & Ikeda, 2020; Patel & Clark-Ginsberg, 2020). A large-scale and nationally representative study by Li and Wang (2020) found an elevated prevalence of loneliness (35.86%) during the COVID-19 pandemic. This finding is similar to those of longitudinal data from 2221 adults in the UK, which indicate that the prevalence of loneliness in the UK has more than doubled during the COVID-19 lockdown (Mental Health Foundation, 2020). The age group that this study finds has been most impacted is young adults, with 44% of individuals aged 18–24 reporting having felt lonely during the first UK lockdown period and nearly half reporting concerns about those feelings.

Whilst transient loneliness can result in emotional distress associated with social disconnection, it is commonplace and adaptive, as it motivates the creation and maintenance of social connections (Cacioppo, Hughes, Waite, Hawkley, & Thisted, 2006). However, when loneliness becomes a chronic and more persistent state, related to a lack of satisfying social relationships over an extended period of time, it triggers neurobiological and behavioural mechanisms that contribute to adverse health consequences (Cacioppo & Hawkley, 2009; McDade, Hawkley, & Cacioppo, 2006).

Cacioppo and Hawkley (2009) theorise a self-reinforcing loop to explain the formation and maintenance of chronic loneliness. They propose that loneliness can increase hypervigilance and cognitive biases towards social threat, leading lonely individuals to anticipate negative social interactions and remember more negative social information (Cacioppo et al., 2016). As a result, lonely individuals may exhibit hostile or pessimistic behaviours which elicit exactly the unwanted responses from others that confirm their negative expectations. Käll, Shafran, Lindegaard, Bennett, Cooper, Coughtry, and Andersson (2020b) state that the challenge of social interaction may also be compounded by individual difficulties such as mental health issues or mobility difficulties. The overall consequence is that a negative self-image is established, along with a desire to avoid social contact, resulting in chronic feelings of loneliness.

Chronic loneliness is emerging as a serious global health concern, as it is a risk factor for a myriad of both physical and mental health conditions. Holt-Lunstad, Smith, Baker, Harris, and Stephenson's (2015) meta-analytic review, which analysed data from 70 independent studies with 3,407,134 participants, found that loneliness increased the likelihood of mortality by 26% even after controlling for multiple covariates. This means that loneliness rivals well-established morbidity risk factors such as physical inactivity, smoking and obesity (Holt-Lunstad et al., 2015). There are also significant health implications for loneliness on young people. Qualter et al. (2013) found that young people with moderate and rising levels or high levels of loneliness had lower perceived general health at age 17 and a higher frequency of visits to their doctor. Furthermore, decreased employee health caused by loneliness has major economic consequences, costing UK employers alone an estimated £2.5 billion per year (Abdallah, Jeffrey, & Michaelson, 2017).

Importantly, loneliness can be viewed as a transdiagnostic construct (Käll, Shafran, Lindegaard, Bennett, Cooper, Coughtry, and Andersson, 2020b) that can occur alongside, as well as predict and exacerbate, a range of mental health conditions (Hawkley & Cacioppo, 2010; Meltzer et al., 2013) including social anxiety (Lim, Rodebaugh, Zychur, & Gleason, 2016), depression (Cacioppo, Hawkley, & Thisted, 2010; Vanhalst et al., 2012), eating disorders (Levine, 2012) and both suicidal ideation and suicidal action (Mezuk, Rock, Lohman, & Choi, 2014; Stickley & Koyanagi, 2016). A cross-sectional study of 7461 adults by Meltzer et al. (2013) found that the likelihood of being lonely is eight

times greater in individuals with a diagnosed mental health difficulty. Additionally, these odds are increased 20-fold for those with two or three mental health diagnoses (Meltzer et al., 2013). Furthermore, an umbrella review of 14 systematic reviews and meta-analyses that reported on 18 outcomes, 795 studies and 746,706 participants (Solmi et al., 2020) found a longitudinal association between loneliness and suicidal action as well as an association between loneliness and depressive symptoms.

A sample of 594 primary care patients showed that loneliness, when left untreated, can independently predict worse anxiety and depression symptoms after one year (van Beljouw, Verhaak, Cuijpers, van Marwijk, & Penninx, 2010). When in treatment, a longitudinal analysis of older adults has found that higher loneliness scores are associated with poorer mental health treatment outcomes (Holvast et al., 2015). In addition, a rapid review of 63 studies and 51,576 children with good mental health found that loneliness significantly increased the risk of depression and anxiety at the time in which loneliness was measured and also nine years later (Loades et al., 2020).

The picture presented by existing research into chronic loneliness highlights that it not only precipitates and exacerbates other serious conditions but is also a distressing psychological phenomenon in its own right that necessitates intervention. Despite this, the development and dissemination of evidence-based interventions for loneliness is still in its infancy compared with interventions for specific mental health disorders (Mann et al., 2017).

A number of systematic reviews have attempted to synthesise the results of loneliness interventions but with significant limitations in several areas. Firstly, nearly all have focused exclusively on interventions for older adults (Cattan, White, Bond, & Learmouth, 2005; Cohen-Mansfield & Perach, 2015; Dickens, Richards, Greaves, & Campbell, 2011; Findlay, 2003; Hagan, Manktelow, Taylor, & Mallett, 2014) rather than on interventions for individuals who have been assessed to be lonely or self-reported as such across the lifespan (Dickens et al., 2011). This limitation is important as loneliness is present and problematic across age groups. Secondly, the majority of reviews have not focused primarily on loneliness, but instead have included studies targeting social isolation. This is problematic as loneliness and social isolation are distinct and only weakly correlated (Coyle & Dugan, 2012); increasing social contact does not necessarily address the perceptual and cognitive components of loneliness. Thirdly, the reviews have been unable to provide conclusive results or robust recommendations due to the heterogeneity of their inclusion criteria and therefore of the types of studies they have included. Taking these various limitations into account, there is a need for additional research that can assess the effectiveness of interventions for loneliness across the lifespan, focus on interventions primarily intended for loneliness, and do this in spite of heterogeneity.

Meta-analysis has the key benefit of providing clearer answers when individual studies are heterogeneous and inconsistent (Haidich, 2010). The first meta-analysis of loneliness interventions was conducted by Masi, Chen, Hawkley, and Cacioppo (2011) who evaluated interventions across the lifespan based on four strategies for reducing loneliness: (a) enhancing social skills; (b) providing social support; (c) increasing opportunities for social interaction; and (d) addressing maladaptive social cognition (biases in attention and cognition towards negative aspects of the social context). Masi et al. (2011) were able to establish a key finding: interventions that target maladaptive social cognitions have the greatest average effectiveness. However, because the finding was based on only four RCTs of social-cognitive interventions, the researchers concluded that it should be independently replicated in order to be considered empirically supported (Masi et al., 2011).

Recently, the first meta-analysis evaluating the effectiveness of a range of interventions to reduce loneliness in children and adolescents was conducted (Eccles & Qualter, 2020). Of the studies included, 25 were RCTs and 14 were single group. Overall, it was found that youth loneliness could be reduced by interventions, although many of the

studies did not target youth for whom loneliness was a clinical or chronic problem and moderator analyses did not ascertain which type of intervention was most effective. However, effect sizes revealed that the type of interventions with the most promise were psychological interventions, as well as social and emotional skills training.

Together, the findings of Masi et al. (2011) and Eccles and Qualter (2020) provide evidence that psychological interventions have promising efficacy for reducing levels of loneliness. This is despite psychological interventions being guided by a broad array of theoretical underpinnings, some of which are not specific to understanding the processes underpinning loneliness but instead target transdiagnostic processes, such as avoidance and cognitive biases. The effectiveness of psychological interventions for loneliness can be understood then by considering the overlap in maintaining mechanisms between loneliness and mental health difficulties that interventions such as CBT were designed to treat (Mann et al., 2017).

Other systematic reviews and meta-analyses have investigated which characteristics make interventions for reducing loneliness effective. Results, however, have been mixed and inconclusive. Various moderating factors have been examined, including: (a) study quality (Cattan et al., 2005; Cohen-Mansfield & Perach, 2015; Eccles & Qualter, 2020); (b) group or individual delivery (Cattan et al., 2005; Eccles & Qualter, 2020; Findlay, 2003; Masi et al., 2011); (c) use of technology in interventions (Chen & Schulz, 2016; Choi, Kong, & Jung, 2012; Cohen-Mansfield & Perach, 2015; Eccles & Qualter, 2020; Poscia et al., 2018; Shah, Nogueras, van Woerden, & Kiparoglou, 2019); and (d) type of intervention (Cohen-Mansfield & Perach, 2015; Eccles & Qualter, 2020; Gardiner, Geldenhuys, & Gott, 2018; Jarvis, Padmanabhanunni, Balakrishna, & Chipps, 2020; Masi et al., 2011; Perese & Wolf, 2005). Two consistent findings from moderator analyses are that technological interventions and interventions with a focus on social cognition display the most potential in reducing loneliness.

Overall, psychological interventions show considerable promise for alleviating chronic loneliness. In spite of this promise, there has been no systematic review or meta-analysis of the effectiveness of psychological interventions for loneliness across the lifespan. Consequently, a synthesis of this type is now needed. This systematic review and meta-analysis advances previous research in multiple distinct ways. Firstly, the review includes only studies that employed randomized controlled trial designs, these being the gold standard due to their potential to eliminate bias in assigning treatments and minimise confounding variables (e.g. Simon, 2001). Secondly, it includes only psychological interventions on the basis that published reviews indicate their promising efficacy for reducing clinical levels of loneliness. Furthermore, psychological treatments for loneliness may also have the added benefit of reducing mental health problems, which often co-occur with loneliness. Thirdly, it only includes studies in which loneliness is the primary or part of the primary target of the intervention. Fourthly it extends the literature search by a further eleven years – the original search carried out by Masi et al. (2011) in 2009 – and into a period of increased research activity with larger and higher quality studies testing interventions for loneliness. And lastly, via moderation and sub-group analysis, it establishes key criteria for intervention success which were not conclusively established in previous research. The moderators investigated are: the type of psychological intervention, the age of participants, the risk of bias rating and the percentage of female participants.

Type of psychological intervention will be included in moderator analysis so that the most effective interventions can be identified and applied in clinical practice. With this being the first systematic review or meta-analysis of psychological interventions for loneliness across the lifespan, it provides the opportunity to establish if age will be a moderator of intervention effectiveness. Due to psychological interventions for loneliness being a relatively novel area it is likely that the studies will vary in quality. Therefore, subgroup analysis will consider if studies with high, medium or low risk of bias are more effective.

Additionally, previous research has been conflicted regarding the link between gender and loneliness (Barreto et al., 2020; Solmi et al., 2020), therefore this will also be examined.

Therefore, the aims of the review are to:

1. Summarise and synthesise the findings of RCTs to address psychological interventions for loneliness across the lifespan.
2. Ascertain the overall effectiveness of psychological interventions compared to control conditions and.
3. Explore the heterogeneity of the interventions and assess whether there were significant moderators of change.

## 2. Methods

The conduct and reporting of the systematic review and meta-analysis follows the guidance of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher, Liberati, Tetzlaff, & Altman, 2009) and the Cochrane Handbook (Higgins et al., 2019). The protocol for the review was registered on the 10th of June 2019 with the PROSPERO database ([www.crd.york.ac.uk/prosperto/](http://www.crd.york.ac.uk/prosperto/)), an international prospective register of systematic reviews. Its registration ID is PROSPERO 2019 CRD42019153376.

### 2.1. Systematic literature search

Search terms were developed in order to identify studies which assessed the effectiveness of psychological interventions in reducing loneliness. These terms were searched in the Ovid Embase, Ovid Medline, PsycINFO, Web of Science and CINAHL databases in November 2019 and updated in November 2020. The key search terms used to identify articles are listed in Table 1. Ovid Embase, Ovid Medline and PsycINFO also allowed the search to include Medical (MeSH) terms which could be ‘exploded’, meaning that the search retrieved all references indexed to that term as well as all references indexed to any narrower term. Additionally, randomized controlled trial filters were added.

### 2.2. Eligibility criteria

The review identified studies reporting quantitative data from randomized controlled trials comparing the effectiveness of psychological interventions to control groups for alleviating loneliness. The search included all published articles up until the end of November 2020.

The inclusion criteria were: (a) peer-reviewed as identified by the journal; (b) a quantitative methodology; (c) an RCT design; (d) loneliness as a primary outcome or part of the primary construct; (e) a psychological intervention based on a psychological theory; (f) available in the English language; (g) published from the year 2000 onwards.

The rationale for including studies published from the year 2000 onwards was that this would reduce overlap with systematic reviews carried out earlier. The additional criterion for inclusion in the meta-analysis was that studies: (h) reported standard quantitative information (mean, standard deviation and sample size) or their authors could provide it when contacted.

**Table 1**  
Search terms.

| Concept                     | Search terms                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Loneliness                  | Lonel* or social isolat*                                                                                                                                           |
| Psychological Interventions | Psychological intervention* or CBT or Cognitive Behavioural Therap* or therap* or IPT or interpersonal therapy or psychotherap* or psychodynamic* or intervention* |

### 2.3. Data collection

Articles were identified, screened and assessed following PRISMA's guidelines (Moher et al., 2009). (See Fig. 1 for flowchart.) Repeat listings of papers across the databases were deleted by the primary researcher. Endnote was used as the systematic review tool. Criteria for inclusion were applied by two independent reviewers (the primary researcher and a PhD Clinical Psychology student with expertise in loneliness) who each examined the abstracts of all 3973 obtained publications. The inter-rater agreement was 97.2% at the abstract screening stage. Conflicts of opinion regarding the eligibility of studies were discussed until consensus was reached.

Following the screening stage, which err on the side of inclusion, 107 papers appeared to meet the eligibility criteria. Four could not be accessed and, because their authors did not respond to an email request for a copy to be supplied, were excluded on the basis that they could not receive a full text screening.

Of the 103 papers that had their full text reviewed, the inter-rater agreement was 81.55%. Any conflicts of opinion regarding inclusion of articles were discussed, with a referral to a third reviewer (RS) if necessary, until consensus was reached. Following full text screening, it was decided that 31 papers met the eligibility criteria and would be included in the systematic review. The decision regarding inclusion in the meta-analysis was made following data extraction.

### 2.4. Data extraction

A headed table was used to guide the extraction of information from the texts. Extraction was initially conducted by the primary researcher. In order to minimise the probability of errors, an independent second coder repeated the data extraction of all quantitative data (Horton et al.,

2010).

Several socio-demographic and clinical characteristics were extracted from the eligible studies including: (a) mean participant age; (b) gender composition; (c) country; (d) population; (e) sample size; and (f) measure of loneliness. Further information was extracted in relation to the psychological intervention: (a) intervention format; (b) type of control group; (c) theoretical model underpinning the intervention; and (d) reported effectiveness of the intervention at reducing loneliness.

The mean, standard deviation and number of participants in the control and intervention group at pre, post and follow up were extracted in order to enable a meta-analysis of the effectiveness of psychological interventions. Authors of the five papers that did not include the necessary statistics for meta-analysis were requested via email to provide these. Two authors did and their papers were included. The other three did not respond and their studies were excluded from the meta-analysis, though not from the systematic review.

### 2.5. Assessment of risk of bias

The risk of bias tool (RoB tool) (Higgins & Altman, 2008) was used to appraise the included studies' quality and potential bias. The RoB tool was chosen based on literature indicating it to be a comprehensive and widely used tool for evaluating RCTs (Higgins et al., 2011; Farrah, Young, Tunis, & Zhao, 2019). This was administered in accordance with the Cochrane Handbook (Higgins, Thomas, & Chandler, 2019). The following five domains were considered in relation to each paper: (a) sequence generation; (b) allocation concealment; (c) blinding of participants, personnel and outcome assessors for each outcome; (d) incomplete outcome data; and (e) selective outcome reporting.

Assessing each domain involved the application of several criteria. The ratings produced by the criteria informed an algorithm which led to

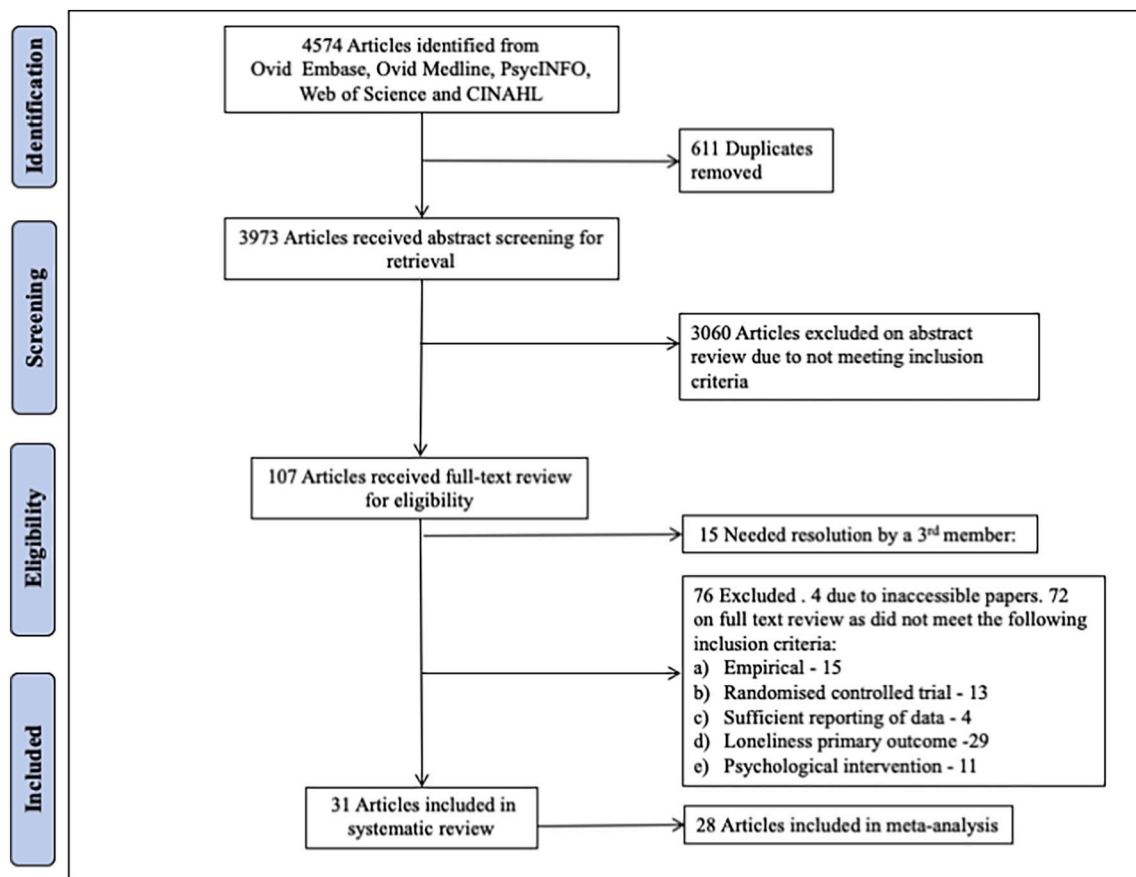


Fig. 1. PRISMA flowchart for the selection process of studies in the systematic review and meta-analysis.



a risk of bias judgement for each domain at one of three levels:

1. Low risk of bias
2. Some concerns
3. High risk of bias.

The domain ratings were then used to inform the overall risk rating for each paper. The primary researcher assessed 25 articles, the second rater (AK) independently assessed 15 articles, nine of which were coded by both authors (29%) independently. Of those nine, ratings were compared and any disagreements resolved by discussion to reach a consensus.

## 2.6. Data synthesis and analysis

All studies included in the systematic review were synthesised and summarised narratively. The meta-analysis was conducted using the software R and the *metafor* package (Viechtbauer, 2010). Standardised mean differences (SMD) were calculated to transform the outcome data into a common metric, thereby enabling the inclusion of other outcome measures within the same synthesis. The SMD were calculated for pre- and post-intervention loneliness scores in the control and intervention groups. The post measurement was taken from the end of the intervention and not from follow up data collection. The difference between the SMD pre to post intervention was calculated in order to account for any baseline difference in loneliness between the groups. The meta-analysis was conducted to ascertain whether the difference from pre to post loneliness in the experimental group was larger than the difference from pre to post in the control group.

Heterogeneity was anticipated due to the range of psychological therapy approaches and study designs used across the eligible studies. Consequently, a random-effects as opposed to a fixed-effect model was used, the former yielding a more conservative estimate and wider confidence interval when there is heterogeneity among effect sizes (Borenstein, Hedges, Higgins, & Rothstein, 2010). REML was used as the method for estimating the heterogeneity variance due to its favourable statistical properties (Langan et al., 2019).

Cochran's Q test and the  $I^2$  statistic were used to assess for heterogeneity in treatment effects. A significant Q statistic indicates varying effect sizes across studies as well as sample or methodological differences that might be causing variance. The  $I^2$  statistic assesses the percentage of variability due to heterogeneity rather than to random error. The  $I^2$  statistic is interpreted as a small (25%), moderate (50%) or high (75%) level of heterogeneity (Higgins, Thompson, Deeks, & Altman, 2003).

To explore possible sources of heterogeneity, meta-regressions and subgroup analyses were conducted to evaluate potential moderators, including age of participant, type of psychological intervention and risk of bias rating. As it was assumed in this case that study variables accounted for some heterogeneity but that there was residual heterogeneity which needed to be accounted for, random effects meta-regression was undertaken.

Additionally, forest plots were created to visually illustrate effect sizes, confidence intervals and outliers. Sensitivity analyses assessed for publication bias through assessing funnel plots of standardised mean differences against standard error.

## 3. Results

### 3.1. Study characteristics

Thirty-one studies were identified for inclusion in the review. Table 2 provides an overview of the studies' characteristics and main findings. All were published between 2003 and 2020. Thirteen were carried out in the USA, three in Iran, two in China, Taiwan and the Netherlands, and one in each of the following countries: Sweden, South Africa, Australia,

Japan, Palestine, Israel, United Kingdom, Canada and Italy. Most of the studies did not report participants' ethnicity.

All studies were randomized controlled trials (RCTs) although some were pilot RCTs. Studies were identified as pilot RCTs based on the explicit description of the RCT being a 'pilot' study within the original paper's methods section. The total number of participants across all studies was 3959. Sample sizes at baseline ranged from 17 to 817 ( $M = 127.71$ ). However, there was often significant attrition of participants. The drop-out percentage based on missing data at post-treatment from baseline to post intervention ranged from 0% to 45.4% ( $M = 10.45\%$ ). Nineteen studies also collected follow-up data beyond the RCT time frame, the follow ups taking place between 1.5 months and 6 months post intervention ( $M = 3.92$ ).

The average age of participants ranged from eight years to 81 years ( $M = 45.20$ ). Four studies were with children, six with young adults (below 25), ten with middle age adults (26–64), five with old adults (65–74) and four with older adults (75+). Five of the studies had samples that were all female and one was conducted with men only. The average percentage of females across all studies was 62.47%. When the studies with single sex samples were removed, the average percentage of females was 57.47%.

The interventions drew on a range of theoretical models: nine used cognitive behavioural therapy techniques, six were integrative, three were mindfulness-based, three were social skills training programmes, one was an interpersonal therapy programme, one was a gratitude intervention, one was a social identity intervention and one was based on reminiscence therapy.

Sixteen of the interventions were group-based, eight were individual and seven were a combination of group and individual. Twenty-four of the interventions were face-to-face and seven were delivered over the phone or via the internet. Fourteen studies used a waitlist control group and participants allocated to this group received the intervention once the intervention group had completed treatment. Eleven studies had active control groups and six offered no treatment to the control group.

Psychological treatments lasted between five days and 52 weeks ( $M = 10.11$  weeks) and sessions were mostly delivered weekly. The mean number of sessions delivered was 9.94, with sessions typically lasting one to two hours, with group treatment sessions on average lasting longer than individual sessions.

The measure used by nineteen studies was either the 20-item, ten-item or eight-item version of the UCLA loneliness scale (Russell, 1996). Four used the De Jong-Gierveld Loneliness Scale (de Jong-Gierveld & Kamphuis, 1985), two used the Illinois Loneliness Questionnaire (ILQ: Asher, Hymel, & Renshaw, 1984), one used the Chinese College Student Loneliness Scale (Li, Jiang, & Li, 2006), one used the Social and Emotional Loneliness Scale for Adults (SELSA: DiTommaso & Spinner, 1993), and one used the Patient-reported Outcomes Measurement Information System (PROMIS) (Hahn et al., 2010).

### 3.2. Quality appraisal

Nine studies were rated as having a low risk of bias, twelve as having some concerns and ten as having a high risk of bias. The most common causes of bias were a lack of blinding personnel and selective reporting of outcomes (See Fig. 2). However, the ratings for selective reporting of outcomes should be interpreted with caution, as study protocols were not available for many studies. These studies were therefore rated as having no information, thus lowering their selective reporting scores.

### 3.3. Meta-analysis

28 studies ( $N = 3039$ ) were included in a meta-analysis of pre- to post-treatment effect sizes (ESs). Psychological interventions significantly reduced loneliness scores compared to control groups ( $p < 0.001$ ). The meta-analysis yielded a small to medium effect favouring the intervention group (overall  $ES\ g = 0.43$ , 95%  $CI = 0.18–0.68$ ). ESs for

**Table 2**

Characteristics of studies.

| Author, Year, Country, Risk of Bias (ROB)                          | Participants                                                                                            | Sample Size                                 | Control Group                                           | Format of Intervention                                                                                                     | Measure of Loneliness                                                                                               | Psychological Theory                                                                                                                                                                                                                                                                                              | Effectiveness Results                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alaviani, Khosravan, Alami, and Moshki (2015)<br>Iran<br>ROB: High | Older women<br><br>Mean age = 67<br>100% female                                                         | 150<br>(I = 75, C = 75)<br>6.7% dropout     | No treatment                                            | Group<br>Face to face<br><br>4 x 60 min sessions, twice per week<br>No follow up                                           | UCLA Loneliness<br><br>Scale – version 3 (20 items)                                                                 | CBT<br>Encourage empowerment in relationships; effective interpersonal interaction; psychoeducation on loneliness. Informed by Social Cognitive Theory                                                                                                                                                            | Intervention led to a significant decrease in loneliness and perceived barriers and increase in perceived social self-efficacy and perceived benefits compared to control<br>Abstract draws a conclusion about improvement which is not evidenced in mean difference     |
| Bartlett and Arpin (2019)<br>US<br>ROB: High                       | Older adults<br><br>Mean age = 73<br>80% female<br>85% Caucasian                                        | 42<br>(I = 23, C = 19)<br>14.3% dropout     | No treatment                                            | Individual<br>Face to face<br><br>21 x daily sessions over three weeks<br>No follow up                                     | Taken from the PANAS (Crawford & Henry, 2004): daily loneliness was assessed with the single negative mood item     | Gratitude<br>Gratitude writing exercise                                                                                                                                                                                                                                                                           | No significant condition differences in loneliness at week 4. However, significant condition-by-baseline depression interaction to predict week-4 loneliness<br>Significant decrease in perceived social isolation in intervention group compared to control             |
| Bruehlman-Senecal et al. (2020)<br>US<br>ROB: Medium               | University students<br><br>Mean age = 19<br>59% female                                                  | 221<br>(I = 100, C = 121)<br>5.43% dropout  | Waitlist control                                        | Individual<br><br>Phone app<br>Open access to app over four weeks. Survey at weeks two and four<br>8 week follow up        | UCLA Loneliness<br>Scale – short version (8 items)                                                                  | CBT<br><br>Nod app incorporates positive psychology, mindfulness-based self-compassion and cognitive behavioural skill-building                                                                                                                                                                                   | No significant condition differences in loneliness at week 4. However, significant condition-by-baseline depression interaction to predict week-4 loneliness<br>Significant decrease in perceived social isolation in intervention group compared to control             |
| Cacioppo et al. (2015)<br>US<br><br>ROB: High                      | US Army service personnel<br><br>Mean age = 24<br>3% female                                             | 817<br>(I = 489, C = 328)<br>28.89% dropout | Active control: Afghanistan cultural awareness training | Group<br>Face to face<br>5 x 2 hr daily sessions<br>No follow up                                                           | UCLA Loneliness<br>Scale – short version (8 items)                                                                  | CBT<br>Social resilience training: modifying maladaptive social cognitions; practising new perspectives                                                                                                                                                                                                           | Significant decrease in loneliness in intervention group. However, no significant difference between groups at follow up                                                                                                                                                 |
| Caputi, Cugnata, & Brombin (2020)<br><br>Italy<br>ROB: Medium      | Children<br>Mean age = 10<br><br>48% female                                                             | 210 (I = 105, C = 105)<br>0%                | Active control: physical stories                        | Individual & group<br>Face to face<br><br>5 x weekly sessions<br>2-month follow up                                         | Illinois Loneliness Questionnaire (ILQ: Asher et al., 1984)                                                         | Social skills training<br>Participants read and discussed mentalistic stories, which contain a discrepancy in beliefs/knowledge/points of view between characters and so tap into the concepts of persuasion, misunderstanding, white lie, irony/sarcasm and contrary emotions in order to develop theory of mind | Significant decrease in loneliness in comparison to control. However, results not significant                                                                                                                                                                            |
| Chiang et al. (2010)<br><br>Taiwan<br><br>ROB: High                | Older men living in a nursing home<br><br>Mean age = 77<br>0% female<br>55% illiterate<br>58% unmarried | 92<br>(I = 47, C = 45)<br>29.4% dropout     | Waitlist control                                        | Group<br>Face to face<br><br>8 x 90 min weekly sessions<br>3-month follow up                                               | UCLA Loneliness<br>Scale – version 3 (20 items)                                                                     | Reminiscence<br>Focusing on positive memories                                                                                                                                                                                                                                                                     | Reduction in loneliness in comparison to control. However, results not significant                                                                                                                                                                                       |
| Choi, Pepin, Marti, Stevens, and Bruce (2020)<br>US<br>ROB: Medium | Older adults<br><br>Mean age = 74<br>62% female                                                         | 89 (I = 43, C = 46)<br>9% dropout           | Active control: videoconference friendly visit          | Individual<br><br>Teleconferencing<br>x5 sessions<br>12-week follow up                                                     | 8-item PROMIS Social Isolation Scale (PROMIS-L)                                                                     | CBT<br>Behavioural activation is a brief, structured behavioural approach that aims to increase and reinforce wellness-promoting behaviours                                                                                                                                                                       | Compared with control, intervention group had greater increase in social interaction and satisfaction with social support and decrease in loneliness<br>Significant difference in loneliness at the end of the intervention and at 3-month follow-up compared to control |
| Cohen-Mansfield et al. (2018)<br><br>Israel<br>ROB: Medium         | Older adults<br><br>Mean age = 77<br>81% female                                                         | 89<br>(I = 45, C = 44)<br>16.9% dropout     | No treatment                                            | Group and/or individual<br>Face to face<br><br>Up to 10 individual meetings<br>Up to 7 group sessions<br>3-month follow up | UCLA Loneliness<br>scale – short version (8 items)<br><br>Also asked about the severity and frequency of loneliness | CBT<br>Addressing psychosocial barriers<br><br>Based on the Cohen-Mansfield and Pargura-Gill (2007) model of depression and loneliness                                                                                                                                                                            | Significant difference in loneliness at the end of the intervention and at 3-month follow-up compared to control                                                                                                                                                         |
| Creswell et al. (2012)                                             | Older                                                                                                   | 40                                          | Waitlist control                                        | Group and individual                                                                                                       | UCLA Loneliness                                                                                                     | Mindfulness                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                          |

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Table 2 (continued)

| Author, Year, Country, Risk of Bias (ROB)                            | Participants                                                                                                                                         | Sample Size                                     | Control Group    | Format of Intervention                                                                                                           | Measure of Loneliness                                                                                                                                                           | Psychological Theory                                                                                                                                                                                                                                                     | Effectiveness Results                                                                                                                                                                                                                        |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US<br>ROB: Low                                                       | Adults<br>Mean age = 65<br>80% female<br>64% Caucasian                                                                                               | (I = 20,<br>C = 20)<br>15.0%<br>dropout         |                  | Face to face<br>8 x 120 min weekly group sessions; 1x day-long retreat and 56 x daily 30 min individual practice<br>No follow up | Scale – version 3 (20 items)                                                                                                                                                    | Distance from cognitions relating to social threat/distress and negative affect                                                                                                                                                                                          | Significant decrease in loneliness compared to control                                                                                                                                                                                       |
| Diab, Punamäki, Palosaari, and Qouta (2014)<br>Palestine<br>ROB: Low | Children<br>Mean age = 11<br>49% female                                                                                                              | 482<br>(I = 242,<br>C = 240)<br>0.0%<br>dropout | Waitlist control | Face to face<br>15 participants per group<br>8 x weekly sessions<br>6-month follow up                                            | A questionnaire combining seven items of the Children's Loneliness Scale (Asher & Wheeler, 1985) and eight items of Friendship Qualities Scale (Bukowski, Hoza, & Boivin, 1994) | Integrative<br>The intervention involved a manualised evidence-based approach which aimed to develop coping skills, emotion regulation and empowerment using narrative, imagery and psycho-educational techniques                                                        | The intervention effect was gender-specific as boys' but not girls' loneliness in peer relations decreased in the intervention group and not among controls                                                                                  |
| Frankel et al. (2010)<br>US<br>ROB: High                             | Children with ASD<br>Mean age = 9<br>14% female<br>45% Caucasian<br>IQ above 60                                                                      | 76<br>(I = 46,<br>C = 30)<br>10.5%<br>dropout   | Waitlist control | Group (concurrent parent and child)<br>Face to face<br>12 x weekly 60 min sessions<br>3-month follow up                          | The Illinois Loneliness Questionnaire (20 items)                                                                                                                                | Social Skills<br>Contains modules that teach social etiquette and specific rules of behaviour which are used by the peer group                                                                                                                                           | Children in the intervention condition reported significantly reduced loneliness compared with control                                                                                                                                       |
| Fukui, Koike, Ooba, and Uchitomi (2003)<br>Japan<br>ROB: Medium      | Women with primary breast cancer<br>Mean age = 53<br>100% female                                                                                     | 47<br>(I = 23,<br>C = 24)<br>0.0%<br>dropout    | Waitlist control | Group<br>Face to face<br>6 x 1.5 hours weekly sessions<br>6-month follow up                                                      | UCLA Loneliness Scale – version 3 (20 items)                                                                                                                                    | Integrative<br>Social comparison; reciprocal exchange of support; health education; coping skills; stress management; peer support and social learning<br>Based on Fawzy and Fawzy (1994) structured psychoeducational group intervention model for patients with cancer | No group-by-time interaction was found because the baseline scores of the control and experimental groups were adjusted and the experimental group showed consistently lower scores at all subsequent time points                            |
| Gantman, Kapp, Orenski, and Laugeson (2012)<br>US<br>ROB: Low        | Young adults with high functioning ASD<br>Mean age = 20<br>29% female<br>58% Caucasian<br>IQ above 70                                                | 17<br>(I = 9, C = 8)<br>0.0%<br>dropout         | Waitlist control | Group<br>Face to face<br>14 x weekly 90 min sessions, caregivers attending concurrently<br>No follow up                          | Social and Emotional Loneliness Scale for Adults (SELSA: DiTommaso & Spinner, 1993)                                                                                             | Social Skills<br>UCLA PEERS for Young Adults Programme (Laugeson, Frankel, Gantman, Dillon, & Mogil, 2012): Evidence-based manualised instruction and rehearsal of social skills related to building close relationships                                                 | Self-reported loneliness decreased for the intervention group compared to control. This group also reported increased participation in social activities, reduced romantic loneliness and the development of friendships compared to control |
| Haslam et al. (2019)<br>Australia<br>ROB: Low                        | Adults with social isolation and a mental health diagnosis or symptoms of depression<br>Mean age = 31<br>64% female<br>74% Caucasian                 | 120<br>(I = 66,<br>C = 54)<br>29.2%<br>dropout  | Waitlist control | Group<br>Face to face<br>4 x weekly 60–90 min sessions<br>No follow up                                                           | UCLA Loneliness Scale – short version (8 items)                                                                                                                                 | Social Identity Approach<br>Manualised workbook<br>Social identity approach to health                                                                                                                                                                                    | The intervention produced a greater reduction in loneliness and social anxiety, fewer general practitioner visits at follow-up and a stronger sense of belonging to multiple groups compared to control                                      |
| Heckman et al. (2006)<br>US<br>ROB: Low                              | Older adults living with HIV/AIDS<br>Mean age = 54<br>32% female<br>50% Caucasian<br>85% unemployed<br>49% gay;<br>15% bisexual;<br>36% heterosexual | 90<br>(I = 44,<br>C = 46)<br>11.1%<br>dropout   | Waitlist control | Group<br>Teleconferencing<br>6–8 participants per group (separated by sexuality)<br>12 x 90 min sessions<br>3-month follow up    | UCLA Loneliness Scale (10 Item version)                                                                                                                                         | CBT<br>Improvement of adaptive emotion-focused coping strategies<br>Based on the Transactional Model of Stress of Coping (Folkman & Lazarus, 1984)                                                                                                                       | No effects on loneliness compared to control. Control group reported significant post-intervention reduction in loneliness                                                                                                                   |

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Table 2 (continued)

| Author, Year, Country, Risk of Bias (ROB)                                          | Participants                                                                                         | Sample Size                              | Control Group                                                                                                                           | Format of Intervention                                                                                                                                                                       | Measure of Loneliness                                                                              | Psychological Theory                                                                                                                                                                                            | Effectiveness Results                                                                                                                              |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Jarvis, Padmanabhanunni, and Chipps (2019)<br>South Africa<br>ROB: High            | Older adults<br>Mean age = 75<br>81% female<br>Ethnicity principally Asian Indian<br>Largely widowed | 32<br>(I = 15, C = 17)<br>9.3% dropout   | Active control (routine care): a generic wellness programme for residents                                                               | Individual and group<br>Face to face (individual), Online (group)<br>40 x twice-weekly 90 min sessions over 5 months<br>No follow up                                                         | De Jong Gierveld Loneliness scale (6 items)                                                        | CBT<br>Psychoeducation on maladaptive cognition linked to loneliness; reflection on cognitive distortion; training in use of technology for increasing social interaction                                       | The intervention reduced loneliness compared to controls and this was maintained at follow up                                                      |
| Jing et al. (2018)<br>China<br>ROB: High                                           | Housebound older adults<br>Mean age = 75<br>70% female                                               | 80<br>(I = 40, C = 40)<br>1.3% dropout   | Active control: Baduanjin qigong                                                                                                        | Individual<br>Online/Phone<br>4 x weekly phone check-ins in first month<br>6 x bi-monthly sessions over 3 months, followed by 9 x monthly sessions over 9 months<br>3 and 6-month follow ups | A self-evaluation of their participants' degree of loneliness based on a 3-point Likert-type scale | CBT<br>Challenging negative cognitions                                                                                                                                                                          | Significant improvement for both control and intervention groups, as well as at follow up. Intervention group showed more improvement than control |
| Käll et al. (2020a)<br>Sweden<br>ROB: Low                                          | General population<br>Mean age = 47<br>71% female                                                    | 73<br>(I = 36, C = 37)<br>10% dropout    | Waitlist control                                                                                                                        | Individual<br>Online<br>8-week programme<br>No follow up                                                                                                                                     | Swedish translation of UCLA Loneliness Scale – version 3 (20 items)                                | CBT<br>Cognitions and behaviours associated with loneliness                                                                                                                                                     | Intervention group felt significantly less lonely post-intervention compared to control                                                            |
| Kremers, Steverink, Albersnagel, and Slaets (2006)<br>The Netherlands<br>ROB: High | Older women<br>Mean age = 63<br>100% female                                                          | 142<br>(I = 63, C = 79)<br>16.2% dropout | No treatment                                                                                                                            | Group<br>Face to face<br>8–12 participants per group<br>6 x 2.5 hr weekly sessions<br>6-month follow up                                                                                      | De Jong Gierveld Loneliness scale (11 items)                                                       | CBT<br>Self-management ability: challenging negative thoughts; goal setting<br>Based on Self-Management of Wellbeing Theory (Steverink, Lindenberg, & Slaets, 2005)                                             | No difference in loneliness reduction compared to control                                                                                          |
| Lai, Li, Ou, and Li (2020)<br>Canada<br>ROB: Medium                                | Older adults<br>Immigrant members of the Chinese community<br>Mean age = 81<br>63% female            | 60 (I = 30, C = 30)<br>0% dropout        | Active control: brief telephone calls from the programme coordinator                                                                    | Individual and group<br>Face to face<br>Weekly, over 5 months<br>No follow up                                                                                                                | De Jong Gierveld Loneliness scale (11 items)                                                       | Social identity approach<br>Peer-based social programme based on Dynamic Social Impact Theory                                                                                                                   | The intervention group showed a statistically significant decrease in loneliness compared to control                                               |
| Lindsay, Young, Brown, Smyth, and Creswell (2019)<br>US<br>ROB: Low                | Community adults<br>Mean age = 32<br>67% female<br>53% Caucasian                                     | 94<br>(I = 57, C = 37)<br>1.1% dropout   | Active control: guidance in free reflection, analytic thinking and problem solving with no explicit mindfulness content                 | Individual<br>Smartphone app<br>14 sessions<br>No follow up                                                                                                                                  | UCLA Loneliness Scale – version 3 (20 items)                                                       | Mindfulness<br>Acceptance toward present-moment experiences                                                                                                                                                     | The intervention reduced loneliness significantly compared with control                                                                            |
| Lloyd-Evans et al. (2020)<br>UK<br>ROB: High                                       | Adults with complex depression or anxiety<br>Mean age = 43<br>73% female                             | 40 (I = 30, C = 10)<br>12.5% dropout     | Active control: standard NHS care, involving monthly meetings with a care coordinator and psychological/psychiatric support on referral | Individual and group<br>Face to face<br>Up to x10 hour-long individual sessions and x3 group sessions over 6 months<br>6-month follow up                                                     | De Jong Gierveld Loneliness scale (11 items)                                                       | Social identity approach<br>The Community Navigator programme is a socially-focused approach, focusing on creating social goals and planning towards increasing social involvement in line with personal values | Reduction in loneliness in intervention group compared with control                                                                                |
| Loucks et al. (2020)<br>US<br>ROB: Low                                             | University students<br>Mean age = 20<br>68% female<br>37% BAME                                       | 96 (I = 47, C = 49)<br>13.5% dropout     | Waitlist control                                                                                                                        | Group<br>Face to face<br>Weekly group, plus daily 45-minute meditation for 6 days per week<br>3-month follow up                                                                              | UCLA Loneliness Scale – version 3 (20 items)                                                       | Mindfulness<br>Mindfulness Based Stressed Reduction (MBSR) for college aged students (MB-College) incorporates a traditional MBSR programme with psychoeducation on wellbeing priorities for this demographic   | Impact on loneliness pre to post was pronounced in the intervention group                                                                          |
| Mascaro et al. (2016)<br>US                                                        | Medical students<br>Mean age = 25                                                                    | 32<br>(I = 21, C = 11)                   | Waitlist control                                                                                                                        | Group and Individual<br>Face to face                                                                                                                                                         | UCLA Loneliness Scale – version 3 (20 items)                                                       | Cognitive Based Compassion Training<br>Meditation; compassion-focused attention training;                                                                                                                       | Participants in the intervention group reported decreased depression and                                                                           |

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Table 2 (continued)

| Author, Year, Country, Risk of Bias (ROB)                                | Participants                                                | Sample Size               | Control Group                                                                  | Format of Intervention                                                    | Measure of Loneliness                        | Psychological Theory                                                                                      | Effectiveness Results                                                                                                                                                                        |
|--------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROB: Medium                                                              | 75% female                                                  | 45.8% dropout             |                                                                                | Group: 10 x 1.5 hr weekly sessions<br>Individual: daily 20 min meditation |                                              | analytic approach to challenging automatic thoughts                                                       | loneliness and an increase in compassion compared to control                                                                                                                                 |
| No follow up<br>Matthews et al. (2018)                                   | Adolescents with a diagnosis of ASD                         | 24<br>(I = 12, C = 13)    | Waitlist control                                                               | Group<br>Face to face                                                     | UCLA Loneliness Scale – version 3 (20 items) | Social Skills<br>The PEERS curriculum: manualised intervention teaching personal and friendship skills    | There was a medium reduction in reported loneliness which approached significance as compared with no significant reduction in the control group. This reduction was maintained at follow up |
| US                                                                       | Mean age = 15                                               | 12.5% dropout             |                                                                                | 14 x 90 min weekly sessions                                               |                                              |                                                                                                           |                                                                                                                                                                                              |
| ROB: Medium                                                              | 25% female                                                  |                           |                                                                                | 4-month follow up                                                         |                                              |                                                                                                           |                                                                                                                                                                                              |
| Ransom et al. (2008)                                                     | Adults with a diagnosis of HIV and with depressive symptoms | 79<br>(I = 41, C = 38)    | Active control (routine care): access to services provided by the AIDS Service | Individual<br>Telephone                                                   | UCLA Loneliness Scale (10 item version)      | IPT<br>Psychoeducation and exploration of interpersonal relationships and conflict                        | No significant change in loneliness in the intervention group or control                                                                                                                     |
| US                                                                       | Mean age = 44                                               | 16.5% dropout             |                                                                                | 6 x 50 min sessions                                                       |                                              |                                                                                                           |                                                                                                                                                                                              |
| ROB: Medium                                                              | 34% female                                                  |                           |                                                                                | No follow up                                                              |                                              |                                                                                                           |                                                                                                                                                                                              |
| Tabrizi, Radfar, and Taei (2016)                                         | 61% Caucasian                                               | 81<br>(I = 41, C = 40)    | Active control (routine care): a brochure regarding self-care.                 | Group                                                                     | UCLA Loneliness Scale – version 3 (20 items) | Integrative                                                                                               | Significant reduction in loneliness scores compared to control                                                                                                                               |
| Iran                                                                     | Breast cancer survivors                                     |                           |                                                                                | Face to face                                                              |                                              | Unstructured supportive expressive discussion groups                                                      |                                                                                                                                                                                              |
|                                                                          | Mean age = 48                                               | 0.0% dropout              |                                                                                | 6–8 participants per group                                                |                                              |                                                                                                           |                                                                                                                                                                                              |
| ROB: Low                                                                 | 67% unemployed                                              |                           |                                                                                | 12 x 90 min weekly sessions                                               |                                              |                                                                                                           |                                                                                                                                                                                              |
| Theeke et al. (2016)                                                     | Chronically ill older adults                                | 27<br>(I = 15, C = 12)    | Active control: 5 x 2 hr weekly sessions of educational information on ageing  | Group<br>Face to face                                                     | UCLA Loneliness Scale – version 3 (20 items) | Integrative<br>LISTEN (Theeke & Mallow, 2015): Rethinking the experience of loneliness to enhance meaning | Reduced loneliness compared to control group                                                                                                                                                 |
| US                                                                       | Mean age = 75                                               | 27.0% dropout             |                                                                                | 3–5 participants per group                                                |                                              | Integrates the key concepts of narrative therapy and CBT                                                  |                                                                                                                                                                                              |
| ROB: Medium                                                              | 89% female                                                  |                           |                                                                                | 5 x 2 hr sessions                                                         |                                              |                                                                                                           |                                                                                                                                                                                              |
|                                                                          | 70% lived alone                                             |                           |                                                                                | No follow up                                                              |                                              |                                                                                                           |                                                                                                                                                                                              |
| van Gestel-Timmermans, Brouwers, van Assen, and van Nieuwenhuizen (2012) | Adults with a history of severe mental illness              | 327<br>(I = 166, C = 161) | Waitlist control                                                               | Group                                                                     | De Jong Gierveld Loneliness scale (11 items) | Integrative                                                                                               | The intervention had no significant effect on loneliness                                                                                                                                     |
| The Netherlands                                                          | Mean age = 44                                               |                           |                                                                                | Face to face                                                              |                                              | A standardised manual: a recovery-enhancing peer support programme                                        |                                                                                                                                                                                              |
| ROB: Medium                                                              | 66% female                                                  | 20.5% dropout             |                                                                                | 7 per group                                                               |                                              |                                                                                                           |                                                                                                                                                                                              |
|                                                                          |                                                             |                           |                                                                                | 12 x 2 hr weekly sessions                                                 |                                              |                                                                                                           |                                                                                                                                                                                              |
|                                                                          |                                                             |                           |                                                                                | 3 and 6-month follow ups                                                  |                                              |                                                                                                           |                                                                                                                                                                                              |
| Zare et al. (2017)                                                       | Mothers of children with cerebral palsy                     | 72<br>(I = 36, C = 36)    | No treatment                                                                   | Individual and group                                                      | UCLA Loneliness Scale (10 item version)      | Integrative                                                                                               | Greater significant improvement for intervention group than control                                                                                                                          |
| Iran                                                                     | Mean age = 28                                               | 0.0% dropout              |                                                                                | Face to face                                                              |                                              | Education through skills training, self-management empowerment and knowledge improvement                  |                                                                                                                                                                                              |
| ROB: Medium                                                              |                                                             |                           |                                                                                | 5 x group sessions                                                        |                                              |                                                                                                           |                                                                                                                                                                                              |
| Zhang, Fan, Huang, and Rodriguez (2018)                                  | University students                                         | 50<br>(I = 34, C = 16)    | No treatment                                                                   | 2 x 1:1 sessions                                                          | Chinese College Student Loneliness Scale     | Mindfulness based Cognitive Therapy                                                                       | Reduction in loneliness compared to control group                                                                                                                                            |
|                                                                          |                                                             |                           |                                                                                | 1.5 month-follow up                                                       |                                              | Maladaptive cognitive patterns/de-identify with perceived social threat                                   |                                                                                                                                                                                              |
| China                                                                    | Mean age = 20                                               | 14.0% dropout             |                                                                                | Group                                                                     |                                              |                                                                                                           |                                                                                                                                                                                              |
|                                                                          |                                                             |                           |                                                                                | Face to face                                                              |                                              |                                                                                                           |                                                                                                                                                                                              |
| ROB: High                                                                | 58% female                                                  |                           |                                                                                | 8 x 2 hr weekly sessions                                                  |                                              |                                                                                                           |                                                                                                                                                                                              |
|                                                                          |                                                             |                           |                                                                                | 3-month follow up                                                         |                                              |                                                                                                           |                                                                                                                                                                                              |

individual studies ranged from -0.42 to 3.04 and substantial significant heterogeneity was observed ( $T^2 = 0.49$ ,  $Q = 228.60$ ,  $p < 0.001$ ,  $I^2 = 89.55\%$ ). See Fig. 3 for the forest plot.

A funnel plot (see Fig. 4) was created to identify potential publication bias. The funnel plot showed some asymmetry with larger studies having effect sizes closer to zero. However, Egger test (Egger, Davey Smith, Schneider, & Minder, 1997) indicated that there was no significant

evidence of funnel plot asymmetry or publication bias ( $p = 0.19$ ).

### 3.4. Subgroup analysis

To explore possible sources of heterogeneity, sub-group analyses were performed considering type of psychological intervention and risk of bias.

### 3.4.1. Types of psychological intervention

Type of intervention was categorised as CBT-based or not CBT-based. This categorisation was decided by three independent coders, one of whom an expert on CBT (RS), who considered the content of the interventions and the theory behind them. Whether interventions were CBT-based did not significantly influence the loneliness outcome ( $I^2 = 0$ ,  $p = 0.60$ ).

A further analysis grouped interventions into seven therapy categories: CBT (10 studies), gratitude (1), reminiscence (1), mindfulness (4), integrative (6), social skills (3) and social identity approach (3). This coding was decided by two independent raters. The interventions therapy had varying effect sizes (see Fig. 5), and the difference between effects was borderline significant ( $Qb = 11.99$ ,  $df = 6$ ,  $p = 0.06$ ). The reminiscence intervention had the highest effect size, followed by social identity approach interventions then CBT.

### 3.4.2. Risk of bias

A subgroup analysis was conducted to ascertain if there was significant variation in effect sizes between studies of low, medium or high risk of bias. Ten studies had a low risk of bias, ten had a medium risk and eight had a high risk. The heterogeneity between the subgroups was non-significant ( $p = 0.84$ ) and  $I^2 = 0\%$ . Therefore, risk of bias rating was not a moderator of reduction in loneliness. Effect sizes and confidence intervals for each of the different risks were 0.38 (95% CI: 0.09, 0.67) across the low risk of bias studies, 0.33 (95% CI: 0.05, 0.60) and for the high risk of bias it was 0.56 (95% CI: -0.18, 1.32).

### 3.5. Meta-regressions

Meta-regressions were conducted to investigate whether numeric study-level variables including 'age of participants' and 'percentage female' were associated with the effectiveness of psychological interventions for loneliness.

The meta-regression model for age was insignificant ( $Qb = 0.02$ ,  $df = 1$ ,  $p = 0.74$ ), indicating that age was not significantly associated with interventions loneliness scores. Sex of participants, measured by the percentage of female participants in each study, was also a non-significant moderator of reduction in loneliness ( $Qb = 0.17$ ,  $df = 1$ ,  $p = 0.68$ ).

## 4. Discussion

This systematic review and meta-analysis are the first to research the effectiveness of psychological interventions for loneliness across the lifespan. The main finding – that psychological interventions are effective at reducing loneliness compared to control groups – represents a significant advance in loneliness research, building on the limited previous evidence (Barreto et al., 2020; Jarvis et al., 2019; Masi et al., 2011).

This finding is particularly critical given the recent upsurge in loneliness and demand for loneliness interventions caused by the current COVID-19 pandemic (Mental Health Foundation, 2020). The effectiveness of psychological interventions for loneliness is therefore an important finding that should inform policy makers, researchers and clinicians considering the pandemic's broader health implications.

Whilst one of the eligibility criteria was that interventions needed to be psychological and based on psychological theory, the theoretical grounding underpinning these interventions is broad. Furthermore, whilst the psychological theory behind an intervention may be applied to loneliness, this was not the origin of these approaches. Therefore, the interventions were not all designed with loneliness in mind. Therefore, a key question is what are the mechanisms which lead psychological interventions to be successful in reducing loneliness?

It is postulated that psychological interventions are successful at reducing loneliness due to the subjective and perceptual nature of loneliness. It is recognised that increasing the amount of social contact alone does not necessarily address the negative interpersonal thoughts or emotional responses, which can maintain loneliness (Käll, Shafraan, Lindegaard, Bennett, Cooper, Coughtrey, and Andersson, 2020b). As psychological interventions are designed for reducing mental health difficulties such as anxiety or depression, which involve mental processes that can overlap with the cognitive changes linked with loneliness, it is proposed that changing a person's mental processes can lead to a change in social behavior, and reduced loneliness over time (Mann et al., 2017).

The transdiagnostic model of chronic loneliness proposed by Käll, Shafraan, Lindegaard, Bennett, Cooper, Coughtrey, and Andersson (2020b) can also add light to which mechanisms are likely to be addressed in some psychological interventions for loneliness. This model suggests that an interpersonal trigger or context, in addition to a value attributed to the importance and worth of relationships, can lead to a perceived discrepancy between desired and actual social situations. These feelings then lead to negative interpersonal appraisals and emotional responses which can result in counter-productive behavioural and cognitive consequences, such as avoidance, self-focused attention and maladaptive cognitive biases. The overall consequence is that a negative self-image is established, along with a desire to avoid social contact, results in chronic feelings of loneliness. Therefore, the most commonly used psychological intervention for loneliness in this meta-analysis, Cognitive Behavioural Therapy (CBT), targets the perceptual and cognitive biases that result in hypervigilance to negative social information (Cacioppo et al., 2006; Cacioppo & Hawkley, 2009). Accordingly, CBT helps individuals to look for disconfirming evidence to reframe perceptions of loneliness and self-efficacy with the aim of changing behaviours, increasing social connections and decreasing loneliness (Käll, Jägholm, Hesser, Andersson, Mathaldi, Norkvist, Shafraan, and Andersson, 2020a). It would be beneficial to also consider mechanisms for change through qualitative research with individuals

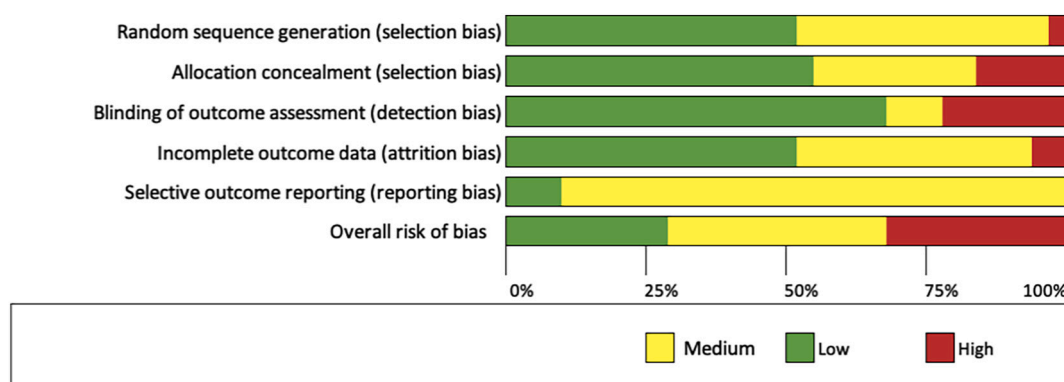


Fig. 2. Risk of bias bar chart.

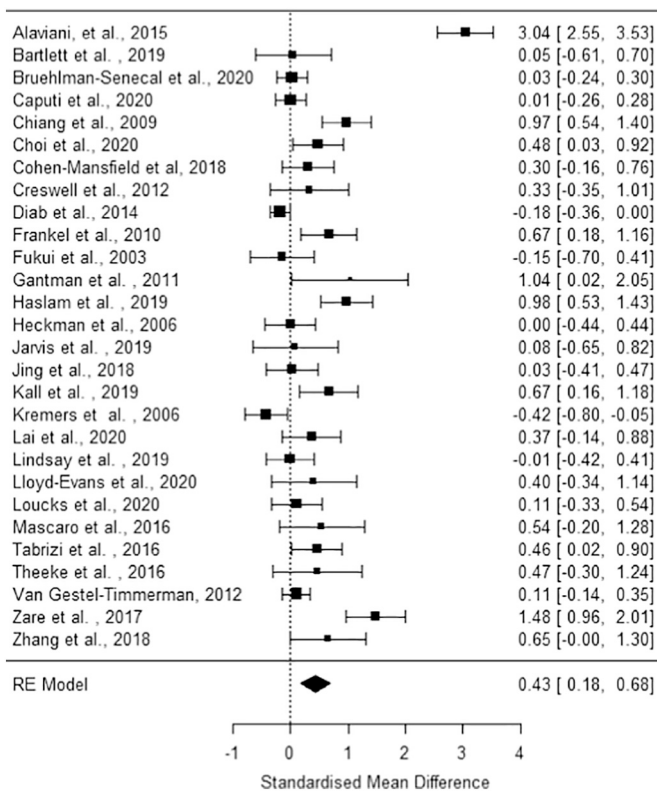


Fig. 3. A forest plot of effect sizes for pre to post treatment.

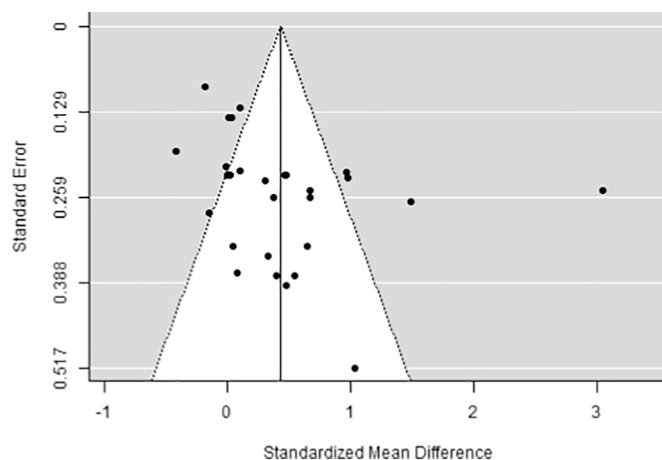


Fig. 4. Funnel plot of meta-analysis.

with lived experience of chronic loneliness, who have undertaken psychological interventions for loneliness, or with mental health practitioners working with lonely individuals (e.g. Stefanidou, Wang, Morant, Lloyd-Evans, & Johnson, 2021). Another key finding was that the effectiveness of psychological interventions varied based on which therapeutic approach was used. Whilst this difference did not reach statistical significance, it indicates that some psychological interventions are better able to alleviate loneliness than others. The reminiscence intervention had the highest effect size, followed by social identity approach interventions and then CBT. However, results should be interpreted with some caution, given that the reminiscence study included was found to have a high risk of bias. Furthermore, due to only having a limited number of studies in most therapy modalities, for example, only one reminiscence based study, further sub-group analyses will need to be conducted as more data and interventions are published.

Interestingly, our subgroup analysis found CBT and social skills had similar effect sizes, differing to Masi et al. (2011) who found cognitive interventions as having the largest effect size and social skills development having no effect. Our finding can be explained by both of these interventions having some overlap despite different theoretical orientations. For example, CBT is often focused on supporting behavioural change such as increased socialising, which will also be a component of social skills interventions.

Sex of participants and targeted age group were not moderators of how effective interventions were. This demonstrates that psychological interventions aimed at all age groups can play an important role in alleviating loneliness for both men and women.

The present systematic review benefits from its methodological rigour, including the use of two independent coders for screening all 3973 abstracts and 103 full texts, with good inter-rater reliability. This minimised the chance of any relevant studies being missed due to human error. The review also utilised a third reviewer when decisions about whether a study met the review's inclusion criteria were unclear.

However, the findings need to be interpreted with an awareness of some limitations. The review only included psychological interventions, making it not possible to compare their efficacy with other types of intervention for loneliness that focus on the wider context of individual's difficulties (e.g. wider community interventions). It has been argued that addressing individuals' maladaptive cognitions prepares them to 'get involved' in their community, although this may have a limited impact if an individual has a lack of connectedness to their community (Mann et al., 2017). Future research should therefore compare the effectiveness of psychological interventions to community interventions or examine whether a combination of a psychological and community-based intervention is more effective than either type alone.

Limitations of some specific studies included in the review include their small sample sizes and lack of underpinning power calculations. Additionally, several studies had very high attrition rates (up to 45.4%) which threatened the validity of their results, especially when the issue of missing data was not analysed further to ascertain if there were differences between those who had completed the intervention and those who had not. In addition, only 61% of studies included a follow up, with the length of follow ups differing, making it difficult to comment on whether the interventions had long-lasting effects.

Whilst some studies targeted loneliness directly and ensured that participants self-reported as feeling lonely as part of their eligibility criteria, other studies did not, instead targeting certain populations that were presumed to be more at risk of loneliness. Moreover, the majority of interventions did not distinguish between transient and chronic loneliness. These findings were also apparent in Eccles and Qualter's (2020) meta-analysis of interventions for lonely young people. Combined, these findings suggest that we need to ask more about loneliness rather than make assumptions about who experiences it.

Future interventions should be designed specifically with loneliness in mind and incorporate the theoretical understanding of the variety of triggers and maintaining factors that exist for chronic loneliness. Additionally, it is important to recognise that lonely individuals are a heterogeneous group and that interventions will need to be tailored to individuals rather than using a 'one-size-fits-all' approach (Perese & Wolf, 2005; Victor, 2018). This level of heterogeneity points to a flexible modular psychological approach being beneficial (Käll, Shafran, Lindgaard, Bennett, Cooper, Coughtrey, and Andersson, 2020b). Additionally, further research should consider which types of psychological intervention are most effective for whom. One way that this question could be addressed is by considering demographic and clinical predictors and moderators of loneliness treatment outcome. Moderators such as intervention length, group vs individual delivery, face to face vs online format should all be investigated further. Finally, future research should assess the long-term benefits of psychological interventions for loneliness and ascertain whether improvements are maintained post-treatment.

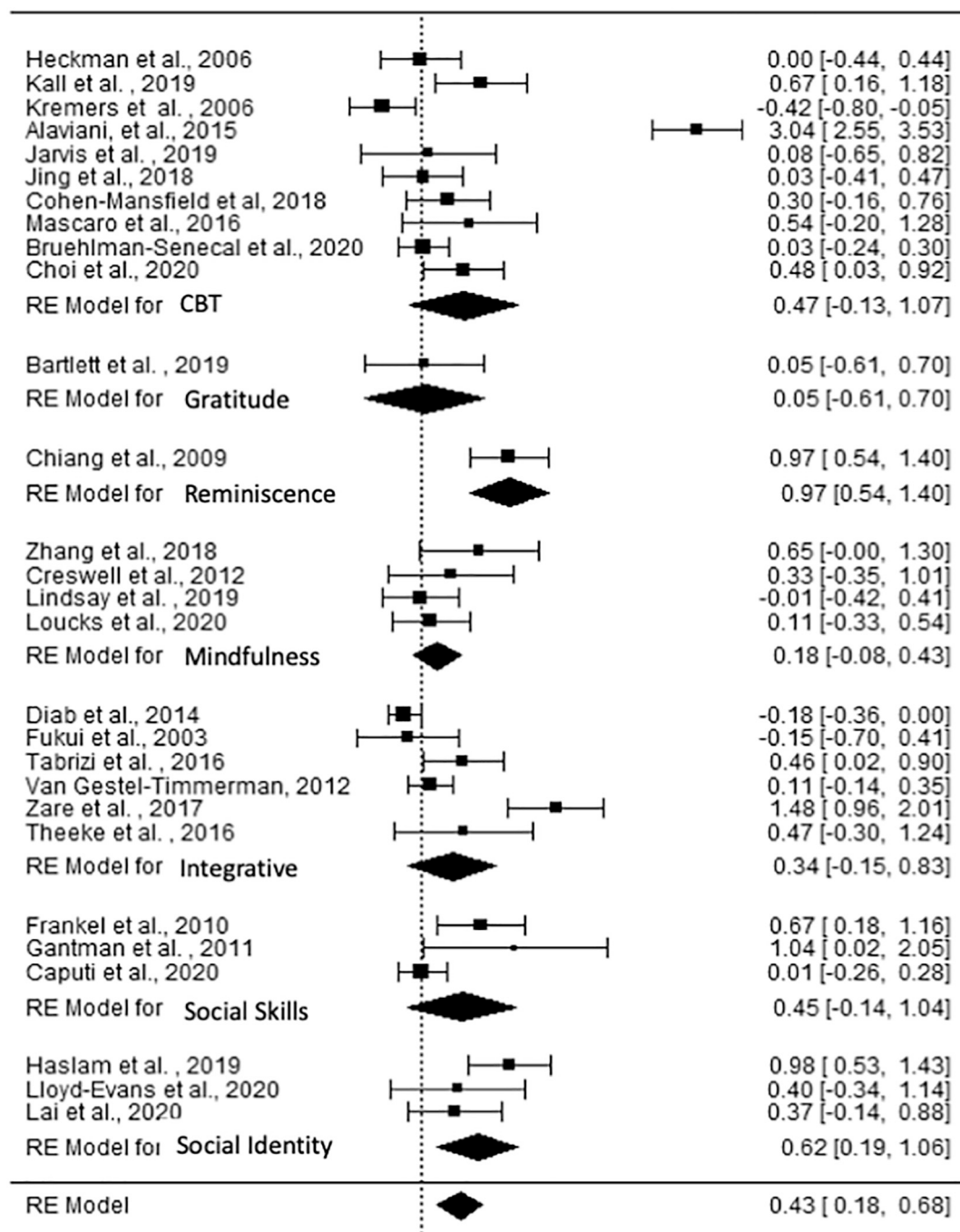


Fig. 5. A forest plot of effect sizes for different types of psychological interventions.

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#### Contributors

NH and RS conceived the idea of this review. NH and GM conducted the literature search; AK was a second reviewer and quality rater. SS assisted with data extraction, coding, checking and referencing. RS was the third reviewer. DL and NH conducted the statistical analysis. NH summarised the findings of previous research studies and wrote the first draft of the manuscript. All authors contributed to, revised and approved the final version.

#### Declaration of Competing Interest

The authors declare that they have no conflicts of interest.

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